Reproducible Data Analysis Workflow

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Titanic — Machine Learning from Disaster dataset

Overview

This project built a complete, notebook-based data workflow in Python for the Titanic passenger dataset. The workflow loads the data, applies cleaning functions, produces exploratory summaries, and generates visual outputs that support an interpretive write-up. The dataset used was *Titanic — Machine Learning from Disaster* from Kaggle (https://www.kaggle.com/competitions/titanic/data). The work was organized in a Jupyter notebook so that code, outputs, and narrative remain in one place, which supports a transparent workflow (Danchev, 2022).

Dataset Description

The source file used in the notebook is the Titanic training dataset, which contains **891 rows** and **12 columns**. Each row represents one passenger and includes a survival outcome (`Survived`), passenger class (`Pclass`), sex, age, family relationship counts (`SibSp`, `Parch`), ticket, fare, port of embarkation, and cabin information. The notebook focused primarily on variables that are both interpretable and relevant to exploratory analysis: `Survived`, `Pclass`, `Sex`, `Age`, `Fare`, `Embarked`, `SibSp`, and `Parch`. After the notebook's cleaning steps, the working dataset contained **889 rows** and **11 columns** because `PassengerId` and `Name` were dropped, `Cabin` was converted to a boolean availability flag, and two rows with remaining missing values were removed.

Workflow Description

Ingestion. The notebook reads the CSV file into a pandas DataFrame and inspects the dataset in tabular form.

Cleaning. Custom helper functions are used to drop unneeded columns and convert sparsely populated fields into boolean indicators. `Cabin` becomes `HasCabin`, `Age` is retained but also converted into `HasAge`, missing age values are filled with `-1`, and remaining rows with missing values are dropped.

Exploratory analysis. The notebook defines reusable functions for summary statistics, grouped or filtered views, and correlation checks. This structure aligns with a reproducible workflow because the same code can be rerun on the same or updated data without manual steps (Danchev, 2022).

Visualizations. The notebook creates a survival count plot, grouped comparisons by class, and a correlation heatmap. Statistical graphics are useful because they make distributional patterns and relationships easier to compare than raw tables alone (Waskom, 2021).

Summary. The final stage is interpretive: the notebook outputs are read as evidence and translated into a narrative about survival patterns, data quality limits, and reproducibility.

Key Decisions and Assumptions

Several design choices shaped the analysis. First, `PassengerId` and `Name` were removed because they function mainly as identifiers or unstructured text and were not central to the questions explored here. Second, `Cabin` was transformed into `HasCabin` because the raw cabin field is highly incomplete; a presence/absence indicator keeps some signal without treating sparse text values as fully usable features. Third, the exploratory analysis focused on survival differences across class, sex, and fare because these variables are directly interpretable in the context of the Titanic dataset and produce clear descriptive contrasts. The plots were designed to show three complementary views: overall survival counts (Figure 1), class-level survival differences (Figure 2), and linear relationships among selected numeric variables (Figure 3). More broadly, organizing the analysis as readable notebook cells with explicit dependencies and explanatory narrative follows recommended good practice for computational notebooks (Rule et al., 2019).

Results and Interpretation

The descriptive results show that the cleaned working dataset has a survival rate of about **38.2%**. Figure 1 shows the overall class imbalance in the target variable: there are more non-survivors than survivors. This matters because even a basic descriptive analysis should recognize that raw counts alone do not imply equal class representation.

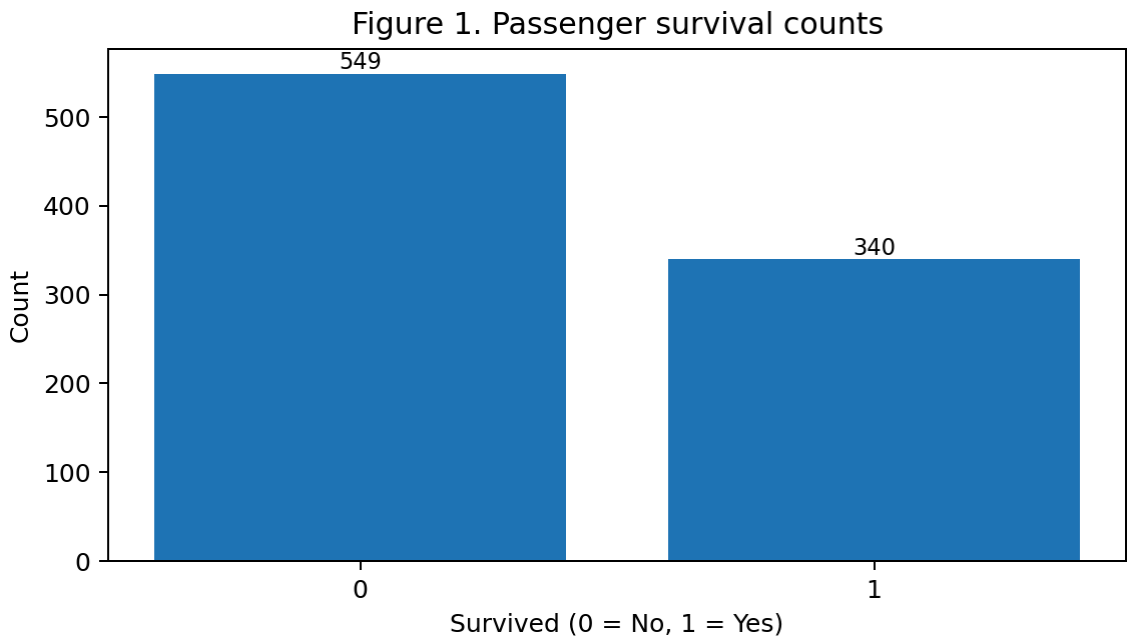

Figure 1. Passenger survival counts from the cleaned working dataset.

Figure 2 highlights the strongest grouped pattern in the notebook: survival differs sharply by passenger class. Mean survival is approximately **0.63** in first class, **0.47** in second class, and **0.24** in third class. The grouped table in the notebook also shows that average fare declines steeply from first class to third class, suggesting that class and fare jointly capture a strong social gradient in the data.

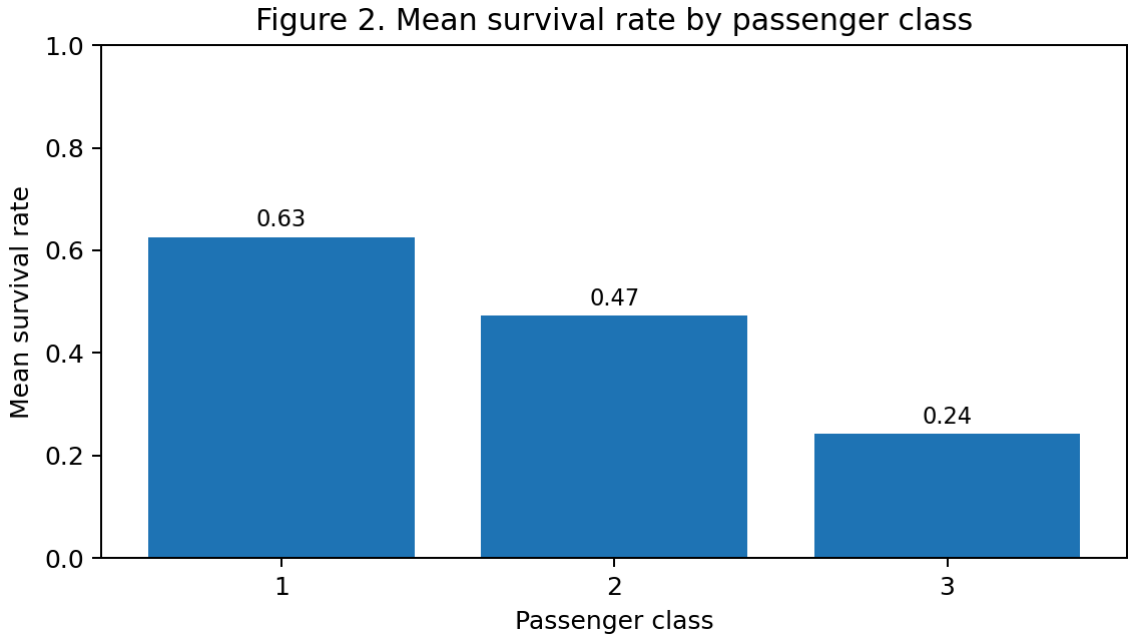

Figure 2. Mean survival rate by passenger class.

The notebook's filtered grouping by embarkation port (`Embarked = 'S'`) adds another important layer: women have markedly higher survival rates than men across all classes, while first-class women have the highest observed survival in that grouped view. Figure 3 shows the selected correlation matrix. The strongest relationship with `Survived` among the plotted numeric variables is a positive correlation with `Fare` (about **0.26**), while `Age` has almost no linear relationship with survival in this cleaned version of the data. That last result should be interpreted cautiously because age was imputed with `-1`, which affects the meaning of the age distribution.

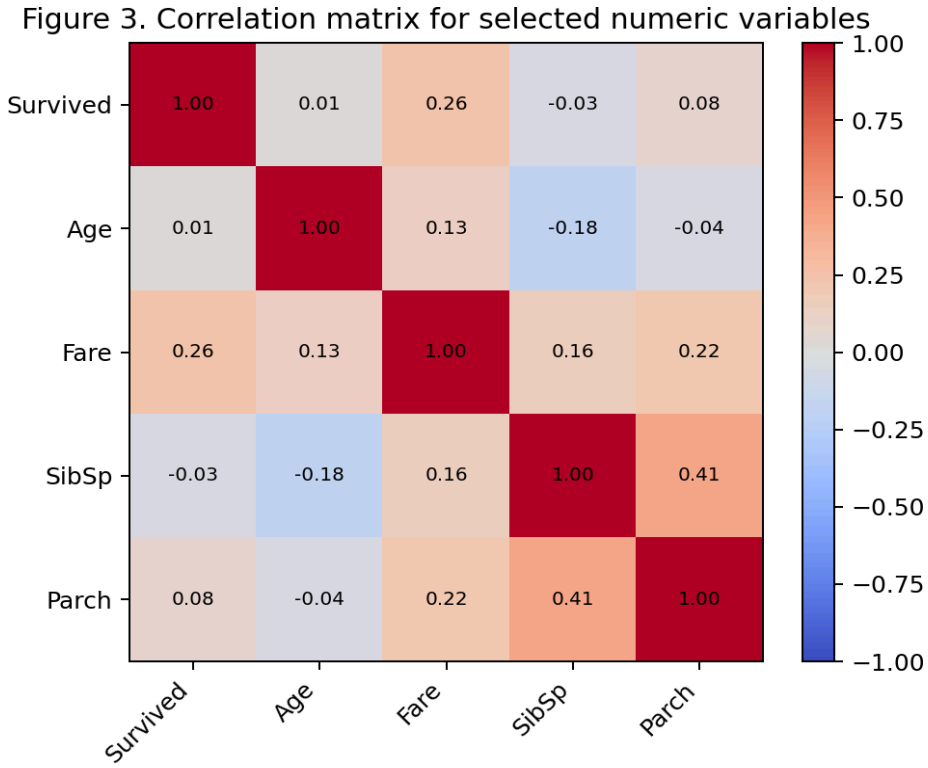

Figure 3. Correlation matrix for selected numeric variables.

Responsible Practice

Cleaning decisions can introduce bias or produce misleading summaries if they are not discussed explicitly. The clearest risk in this notebook is the handling of missing ages: filling missing `Age` values with `-1` preserves a numeric column but introduces an artificial value that lowers the mean age and can distort correlations, distributions, and any downstream interpretation of passenger age. Similarly, collapsing `Cabin` into `HasCabin` removes potentially meaningful **details** such as deck information. A better next step would be to compare multiple missing-data strategies, for example median imputation by subgroup or a separate analysis that excludes imputed age values when producing descriptive summaries. Responsible practice in exploratory work includes making these limitations visible rather than presenting a single cleaned version as if it were neutral (Danchev, 2022).

Reproducibility

Someone else can rerun this work by cloning the repository, creating a virtual environment, installing dependencies with `pip install -r requirements.txt`, and opening `data_analysis_workflow.ipynb` in Jupyter. The dependency file records package versions, which is important because notebook output and plotting behavior can change when packages change; Danchev (2022) specifically notes the value of recording package dependencies for reproducible notebook environments. The project's Git workflow also supports reproducibility because commits provide a history of analytical changes and branching makes it easier to separate experimentation from the main notebook. Together, the notebook, the CSV data file, the requirements file, and version control create a rerunnable workflow rather than a one-off analysis. This is consistent with recommendations to make notebook structure explicit, document dependencies, and keep code and explanation together (Rule et al., 2019).

References

Danchev, V. (2022). *Reproducible Data Science with Python: An Open Learning Resource*. Journal of Open Source Education, 5(56), 156. https://doi.org/10.21105/jose.00156

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